

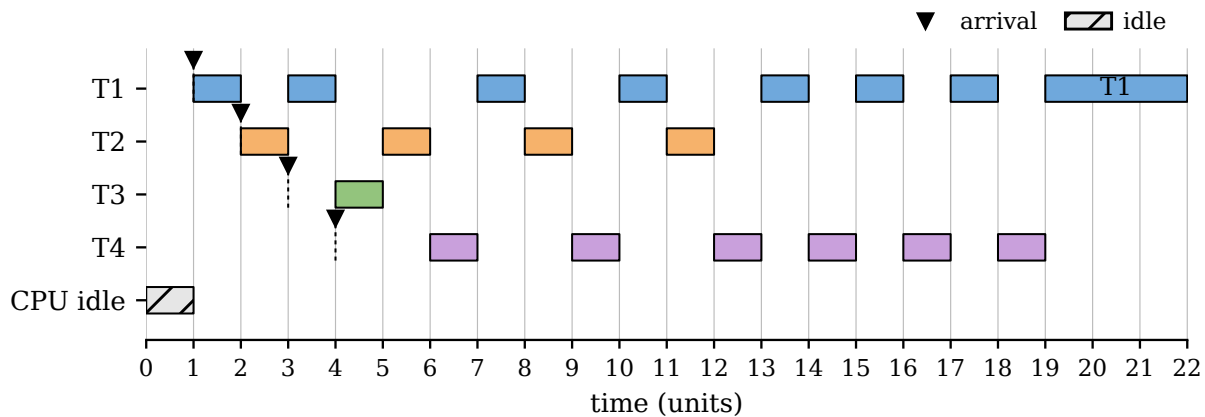
Ex. 2.1 (15 pts) — Schedule diagrams: Round-Robin, FIFO and SJF

Task set (a_i, e_i) : $T_1(1, 10)$, $T_2(2, 4)$, $T_3(3, 1)$, $T_4(4, 6)$.

Conventions.

- Time is discrete: the task dispatched at instant t holds the CPU over $[t, t + 1)$, and a task with arrival time a_i is runnable at instant a_i .
- Nothing has arrived before $t = 1$, so $[0, 1)$ is idle under every policy; it is drawn, not cropped.
- FIFO is non-preemptive and dispatches in arrival order.
- SJF is the classical non-preemptive form: at each dispatch point it picks the arrived, unfinished task with the smallest *total* execution time, and that task then runs to completion.
- Round-Robin uses $q = 1$. Where a task arrives at the same instant that the running task exhausts its quantum, the arriving task is enqueued first and the preempted task then joins the tail of the ready queue.
- Every tie — equal bursts in SJF, equal standing in the ready queue — goes to the smaller index.

Total busy time is $10 + 4 + 1 + 6 = 21$ units, so with the single idle unit all three policies complete at $t = 22$.



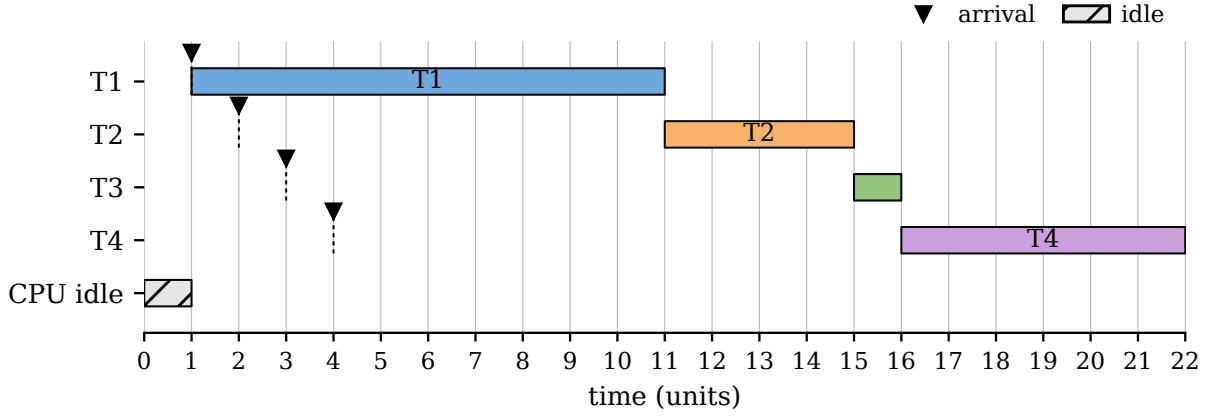


Figure 2: FIFO (non-preemptive).

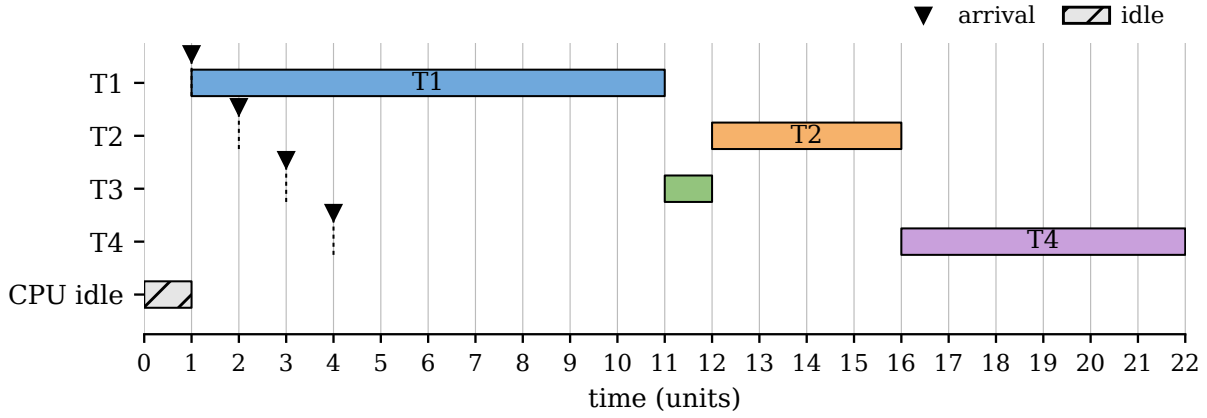


Figure 3: SJF (non-preemptive).

Task	a_i	e_i	Round-Robin ($q = 1$)			FIFO			SJF		
			C	T	W	C	T	W	C	T	W
T_1	1	10	22	21	11	11	10	0	11	10	0
T_2	2	4	12	10	6	15	13	9	16	14	10
T_3	3	1	5	2	1	16	13	12	12	9	8
T_4	4	6	19	15	9	22	18	12	22	18	12
average				12.00	6.75		13.50	8.25		12.75	7.50

Table 1: Completion C , turnaround $T = C - a_i$ and waiting $W = T - e_i$ times.

Round-Robin gives the lowest average turnaround and waiting times because the three short tasks interleave with T_1 instead of queueing behind its 10-unit burst, and FIFO the highest because T_3 waits 12 units to run for 1. The preemptive variant of SJF (SRTF) would give a different schedule: T_2 preempts T_1 at $t = 2$ and T_3 preempts T_2 at $t = 3$.